CO3 AT2 – Case Study

DSA 0401 – Computer Vision with Open cv

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Case Study 1: Defect Detection in Industrial Parts

A. Region-Based Representation and Medial Axis Computation

Region-based representation represents the complete shape of a component as a connected region of pixels. For binary images of gears or machine components, the object can be separated from the background using segmentation. Important properties such as area, perimeter, centroid, bounding box, and connected components can then be calculated.

This helps in defect detection because:

- A missing part reduces the area of the component and changes its boundary.
- A small protrusion increases the area and changes the shape of the boundary.
- Connected-component analysis can identify unwanted or separated regions caused by defects.
- Comparing these measurements with a normal reference component helps classify the component as normal or defective.

Medial axis computation or skeletonization reduces the object to a thin centreline while preserving its basic structure. It represents the shape using its main branches and their connections.

For example:

- A normal gear has a regular and symmetric skeleton.
- A missing tooth may cause a change or interruption in the skeleton.
- An extra protrusion may create an additional branch.

Therefore, region-based representation detects overall shape and area changes, while the medial axis helps identify structural changes and irregularities.

B. Effect of Noise and Preprocessing

Noise introduced during image acquisition can significantly affect shape analysis. Noise may create small unwanted regions, holes, rough boundaries, or false protrusions. As a result, measurements such as area, perimeter, skeleton, and shape descriptors may become inaccurate.

For example, a noisy pixel near the boundary of a gear may appear as a small protrusion and could be incorrectly classified as a defect.

The following preprocessing steps can improve accuracy:

1. Grayscale conversion – Convert the image into grayscale if required.
2. Thresholding – Convert the image into a clean binary image separating the object from the background.
3. Noise filtering – Apply median or Gaussian filtering to reduce unwanted noise.
4. Morphological opening – Removes small unwanted objects and protrusions.
5. Morphological closing – Fills small gaps and holes in the object.
6. Hole filling – Removes unwanted holes inside the component.
7. Connected-component analysis – Removes very small isolated regions.
8. Boundary smoothing – Reduces irregularities in the object boundary before shape analysis.

These preprocessing techniques produce a cleaner binary shape and make defect detection more reliable.

C. Fourier Descriptors vs. Wavelet Descriptors

Fourier descriptors represent the shape using the frequency components of its boundary. The boundary of a component is converted into a mathematical signal, and the Fourier transform is used to obtain descriptors.

Advantages of Fourier descriptors:

- Simple and computationally efficient.
- Good for representing the overall shape of gears and machine parts.
- Relatively insensitive to translation, rotation, and scaling when normalized.

- Effective when defects cause significant changes to the overall boundary.

Limitation: Fourier descriptors may not represent very small or localized defects very effectively because they mainly describe the global shape.

Wavelet descriptors represent shape information at different scales and locations.

Advantages of wavelet descriptors:

- Can capture both global and local shape information.
- Effective for detecting small protrusions, missing teeth, cracks, or irregular boundary regions.
- Provides multi-resolution analysis, allowing defects to be studied at different scales.

Limitation: They are generally more complex than Fourier descriptors and may require more computation and careful selection of wavelet parameters.

Comparison

Feature	**Fourier Descriptors**	**Wavelet Descriptors**
Representation	Frequency-based	Multi-resolution
Global shape	Excellent	Good
Local defects	Moderate	Excellent
Small protrusions	Less effective	More effective
Missing parts	Effective	Highly effective
Computational complexity	Lower	Higher
Suitable for	Overall shape classification	Local and multi-scale defects

Conclusion

For normal vs. defective industrial components, Fourier descriptors are suitable when the defect produces a major change in the overall shape. However, wavelet descriptors are generally more effective when the defects are small or localized, such as small protrusions or missing teeth. Therefore, for this case study, a wavelet-based approach would be preferable for detecting small defects, while Fourier descriptors can be used for fast overall shape classification.

Case Study 2: Medical Image Segmentation

A. Application of Active Contours (Snakes)

Active contours, commonly called snakes, are deformable curves used to detect object boundaries in an image. In CT scans, a contour is initially placed around or near the organ of interest.

The snake then gradually moves toward the actual organ boundary by minimizing an energy function consisting mainly of:

- **Internal energy:** Maintains smoothness and prevents the contour from becoming irregular.
- **External energy:** Attracts the contour toward strong image features such as edges and intensity changes.
- **Image-based energy:** Uses information from the CT image to identify the organ boundary.

The general process is:

1. Obtain the CT image and apply preprocessing such as noise reduction.
2. Place an initial contour near the organ.
3. Allow the contour to deform toward the organ boundary.
4. The contour stops when it reaches a stable boundary.
5. The final contour represents the segmented organ.

Active contours are useful because organ boundaries can be irregular and slightly different from patient to patient.

B. Challenges in Initialization and Parameter Tuning

Accurate segmentation depends strongly on the initial contour and the parameters used by the snake model.

Challenges in initialization:

- If the initial contour is too far from the organ, the snake may not reach the correct boundary.
- If the contour is placed inside the wrong region, it may converge to an incorrect boundary.
- Manual initialization can be time-consuming.
- Different patients have different organ shapes and sizes, making a fixed initial contour unsuitable.

Challenges in parameter tuning:

- Smoothness parameter: Too high may remove important boundary details; too low may produce an irregular contour.
- Edge attraction: If too weak, the contour may fail to reach the organ boundary. If too strong, it may be attracted to unwanted edges.
- Step size: A large value may cause instability, while a small value increases computation time.
- Stopping condition: Incorrect settings may cause premature stopping or unnecessary iterations.

Therefore, preprocessing, automatic initialization, and adaptive parameter selection can improve segmentation accuracy.

C. Level Set Methods for Splitting and Merging Organs

Level set methods represent the contour as a function rather than explicitly representing it as a fixed curve. The contour can automatically expand, shrink, split, or merge during the segmentation process.

This is an important advantage when the shape of organs is complex.

For example:

- If one segmented region needs to split into two regions, the level set can naturally divide the contour.

- If two regions need to **merge into one region**, the level set can combine them.
- It can handle complex and changing organ boundaries without manually modifying the contour.
- It can continue working even when the boundary changes topology.

In contrast, traditional snakes usually have difficulty handling situations where a contour must split or merge because their topology is relatively fixed.

Conclusion

Active contours are effective for segmenting organs by deforming a contour toward their boundaries. However, their performance depends heavily on initialization and parameter selection. Level set methods provide greater flexibility because they can naturally handle changes in topology, including splitting and merging. Therefore, level set methods are particularly suitable for complex medical images where organ boundaries vary significantly.